

Item No.	Component Name	Component Type	Function/Role	Status	Quantity	Unit Price	Total Price	Source	Link	Order Date	Notes/Remarks
1	Raspberry Pi 4	Single-board computer	Central control unit running ROS and interfacing logic	Integrated	1			University			Will be provided by university
2	STM32 F446RE	Microcontroller	Real-time motor control and Modbus communication	Purchased and integrated	1	20,50 €	25,50 €	Reichelt	https://www.reichelt.de		Total price incl. shipping cost
3	DiGiYes MAX485	Transceiver module	Transceiver for RS-485-communication	Purchased and integrated	2		18,77 €	Amazon (Offer for a 10 pack, unable to find lower quantity.)	https://www.amazon.de		Other module possible if characteristics are the same. May be provided by Uni or ATB
4	Jumper Wire	Wire	Electrical connections between components	Available	Various			Students			Will be provided by students
5		Component housing		Manufactured by students, all .stp files in TUBCloud	1			Students			Preferably 3D printed from PETG, as the space requirements are predefined by the existing robot platform.
6	APKLVSr	DC/DC Converter	Convert 24V current from battery to 5V for RPi and STM	Purchased and integrated	1	9,59 €	9,59 €	Amazon	https://www.amazon.de		
7	57BL04	Brushless DC Motor	Drive propulsion (Horizontal axis)	Existing part	4				https://www.mouser.de		Provided/assembled by student in previous project
8	23HE22-2804S	Servo Motor	Steering (Vertical axis)	Existing part	4				https://www.mouser.de		
9	BLD-305S	BLDC Motor Driver	Controls BLDC motor via Modbus	Existing part	4				https://www.mouser.de		
10	MKS Servo57D	Servo Motor Driver	Controls servo motor via Modbus	Existing part	4				https://mouser.de		

Total Price:53,86 €